

eBPF Hook Points for the AMF Threat-Aware State Machine

Mapping the 23-state / 6-endpoint model in `analysis/amf_state_machine.py` onto real free5gc AMF Go functions for uprobe / XDP instrumentation

Module: `github.com/free5gc/amf` · Source tree: `internal/` · Generated 2026-07-16

How to read this document

The state machine in `amf_state_machine.py` is a conceptual, threat-aware decomposition of UE registration — it is **not** the real free5gc FSM (which has only 6 states: see `AMF_UE_State_Verification.pdf`). Several of the 23 states here live *inside* a single real FSM state (e.g. `IDENTITY_PENDING`, `AUTH_VECTOR_PENDING`, `NAS_AUTHENTICATING` are all sub-steps of the real `Authentication` state). For those, the correct eBPF hook is not the FSM state-entry function but the specific sub-function that performs that step — this document gives you that function, so a userspace agent reading the eBPF ring buffer can reconstruct entry into each of the 23 conceptual states/edges by watching for these symbols to fire, and write the resulting state into the eBPF map (`BPF_MAP_TYPE_HASH` keyed by UE identity, e.g. `RAN-UE-NGAP-ID` or `5G-TMSI`).

- UPROBE

function entry, args give context
- URETPROBE

function return, inspect return value/error
- XDP

packet-ingress boundary, below Go runtime
- DERIVED

no single function; correlate multiple probe hits

Part 1 — Normal-Path States (13)

01 REG_RECEIVED			
Type	Symbol	Location	Purpose
XDP	SCTP DATA, PPID=60 (NGAP)	below Go runtime	Raw N2 packet ingress on the SCTP association, before any userspace copy — see Part 3.
UPROBE	internal/ngap.handleInitialUEMessageMain	ngap/handler.go:430	Decodes NGAP <code>InitialUEMessage</code> IEs, creates the <code>RanUe</code> , extracts the embedded NAS PDU.
UPROBE	internal/nas.HandleNAS	nas/handler.go:14	True N2→NAS/GMM entry point: integrity-checks and dispatches the first NAS message. Best single symbol for this state.

02 CONTEXT_LOOKUP			
Type	Symbol	Location	Purpose
UPROBE	internal/gmm.contextTransferFromOldAmf	gmm/handler.go:579	GMM-level decision to contact the old serving AMF (fires when <code>ue.ServingAmfChanged</code>).
URETPROBE	internal/sbi/consumer.(*namfService).UEContextTransferRequest	sbi/consumer/amf_service.go:245	Literal <code>Namf_Communication_UEContextTransfer</code> HTTP call to the old AMF; return value gives the "response" edge.

03 IDENTITY_PENDING			
Type	Symbol	Location	Purpose
UPROBE	internal/gmm/message.SendIdentityRequest	gmm/message/send.go:91	Fires on entering <code>IDENTITY_PENDING</code> — outbound "Request identity".
UPROBE	internal/gmm.HandleIdentityResponse	gmm/handler.go:1536	UE's identity reply — the "Response identity" edge.

04 AUTH_VECTOR_PENDING			
Type	Symbol	Location	Purpose
UPROBE	internal/gmm.AuthenticationProcedure	gmm/handler.go:1662	Orchestrates AUSF selection (NRF lookup) then requests the auth vector.
URETPROBE	internal/sbi/consumer.(*nausfService).SendUEAuthenticationAuthenticateRequest	sbi/consumer/ausf_service.go:51	Literal <code>Nausf_UEAuthentication_Authenticate</code> call; return gives "auth vector response" / "AUSF reject" / timeout.

05 NAS_AUTHENTICATING

Type	Symbol	Location	Purpose
UPROBE	internal/gmm/message.SendAuthenticationRequest	gmm/message/send.go:136	Outbound NAS Authentication Request/challenge to UE — entry into this state.
UPROBE	internal/gmm/HandleAuthenticationResponse	gmm/handler.go:1973	Validates RES*/HRES*; success/fail branches drive AuthSuccessEvent / AuthFailEvent .
UPROBE	internal/gmm/HandleAuthenticationFailure	gmm/handler.go:2124	UE-reported auth failure (MAC failure, sync failure, non-5G-auth-unacceptable).

06 NAS_SECURITY_PENDING

Type	Symbol	Location	Purpose
UPROBE	internal/gmm.SecurityMode	gmm/sm.go:260	Real FSM state-entry function — canonical hook for this state.
UPROBE	internal/gmm/message.SendSecurityModeCommand	gmm/message/send.go:434	Outbound Security Mode Command (called from sm.go:298).
UPROBE	internal/gmm/HandleSecurityModeComplete	gmm/handler.go:2292	UE's Security Mode Complete — "integrity success" edge.
UPROBE	internal/gmm/HandleSecurityModeReject	gmm/handler.go:2354	UE-rejected security mode.

07 UDM_REGISTERING

Type	Symbol	Location	Purpose
UPROBE	internal/gmm.communicateWithUDM	gmm/handler.go:1004	Enclosing GMM step: UDM discovery → UECM registration → SDM Get sequence. Best single symbol for state entry.
URETPROBE	internal/sbi/consumer.(*nudmService).UeCmRegistration	sbi/consumer/udm_service.go:382	Literal Nudm_UECM_Registration call.

08 SUBSCRIPTION_LOADING

Type	Symbol	Location	Purpose
URETPROBE	internal/sbi/consumer.(*nudmService).SDMGetAmData	sbi/consumer/udm_service.go:95	Nudm_SDM_Get for AM data — fires immediately after UeCmRegistration succeeds.
UPROBE	(*nudmService).SDMGetSmfSelectData / SDMGetUeContextInSmfData / SDMSubscribe	udm_service.go:143,187,230	Additional SDM calls in the same sequence — hook for full subscription-loading coverage.

09 POLICY_ASSOCIATING

Type	Symbol	Location	Purpose
URETPROBE	internal/sbi/consumer.(*npcfService).AMPolicyControlCreate	sbi/consumer/pcf_service.go:47	Literal Npcf_AMPolicyControl_Create call, invoked from gmm/handler.go:733 .

10 UE_CONTEXT_READY

Type	Symbol	Location	Purpose
UPROBE	internal/gmm/message.SendRegistrationAccept	gmm/message/send.go:562	No distinct FSM state exists here in real free5gc (still inside ContextSetup); this is the concrete "context ready, Accept being sent" point — final call of HandleInitialRegistration (handler.go:774), fires only after policy association succeeds.

Decides internally (line ~591) whether to escalate to SendInitialContextSetupRequest (non-3GPP) or SendN2Message (3GPP) — see state 12.

11 SM_CONTEXT_PENDING

Type	Symbol	Location	Purpose
UPROBE	internal/gmm.CreatePDUSession	gmm/handler.go:222	GMM-side orchestrator, invoked from HandleULNASTransport on a PDU-session-establishment ULNASTransport.
URETPROBE	internal/sbi/consumer.(*nsmfService).SendCreateSmContextRequest	sbi/consumer/smf_service.go:153	Literal Nsmf_PDUSession_CreateSMContext call (handler.go:274).

12 INITIAL_CONTEXT_SETUP

Type	Symbol	Location	Purpose
UPROBE	internal/ngap/message.SendInitialContextSetupRequest	ngap/message/send.go:462	Outbound NGAP setup request — called from <code>SendRegistrationAccept</code> (non-3GPP) or conditionally from <code>SendN2Message</code> (3GPP, send.go:1344, guarded by <code>!ranUe.InitialContextSetup && (UeContextRequest pduSessionResourceSetupRequestList)</code>).
UPROBE	internal/ngap.handleInitialContextSetupResponseMain	ngap/handler.go:948	Success handler; sets <code>ranUe.InitialContextSetup = true</code> .

13 REGISTERED_CONNECTED

Type	Symbol	Location	Purpose
UPROBE	internal/gmm.Registered	gmm/sm.go:65	Real FSM state-entry function — exact analog of the terminal success state. Entry code increments the UE-connectivity gauge if CM-connected.
UPROBE	internal/gmm.HandleRegistrationComplete	gmm/handler.go:2237	Sends <code>ContextSetupSuccessEvent</code> (line 2283) that drives the FSM transition into <code>Registered</code> .

Part 2 — Failure / Threat States (10)

14 REG_REJECTED

Type	Symbol	Location	Purpose
UPROBE	<code>internal/gmm/message.SendRegistrationReject</code>	<code>gmm/message/send.go:393</code>	Single choke-point for every registration-reject path. Argument <code>cause5GMM</code> classifies the reason. Called from <code>handler.go:648</code> (missing capability), 1159/1165 (NSSF reject), 1685/1701 (AUSF selection fail), <code>sm.go:286</code> (security alg mismatch).

15 AUTH_FAILED

Type	Symbol	Location	Purpose
UPROBE	<code>internal/gmm.Authentication</code> (<code>AuthFailEvent</code> case)	<code>gmm/sm.go:222–233</code>	Actual state-transition handler: sends <code>UEContextReleaseCommand</code> , removes <code>RanUe</code> , calls <code>RemoveAmfUe</code> — this directly realizes the <code>CLEANUP_REQUIRED</code> edge too.
UPROBE	<code>internal/gmm.HandleAuthenticationResponse</code> / <code>HandleAuthenticationFailure</code>	<code>gmm/handler.go:1973 / 2124</code>	RES*/AUSF-failure branches (~2007, ~2043) and MAC/sync-failure branches (2140, 2152, 2179) that fire <code>AuthFailEvent</code> .

16 REPLAY_SUSPECTED

DERIVED SIGNAL

Type	Symbol	Location	Purpose
UPROBE	<code>internal/gmm.HandleAuthenticationFailure</code> (<code>SynchFailure</code> case)	<code>gmm/handler.go:2175–2183</code>	Reads <code>ue.AuthFailureCauseSynchFailureTimes</code> ; escalates to <code>AuthFailEvent</code> at ≥ 2 repeated sync failures — the closest built-in analog of a replay/relay signature.
DERIVED	<code>ue.MacFailed</code>	<code>nas/handler.go:61</code> (set)	No dedicated replay-detection module exists in <code>free5gc</code> . Set at <code>nas/handler.go:61</code> , read at 12+ sites in <code>gmm/handler.go</code> (lines 48, 407, 433, 614, 1568–1650, 2297, 2497). Build <code>REPLAY_SUSPECTED</code> as an eBPF-map counter: repeated <code>MacFailed==true</code> across consecutive <code>HandleNAS</code> hits for the same UE key.

17 SECURITY_FAILED

Type	Symbol	Location	Purpose
UPROBE	<code>internal/nas/nas_security.Decode</code>	<code>nas/nas_security/security.go:120</code>	Low-level MAC/integrity verification routine; returns <code>integrityProtected</code> bool consumed by <code>HandleNAS</code> .
UPROBE	<code>internal/gmm.HandleSecurityModeReject</code>	<code>gmm/handler.go:2354</code>	UE-side security mode rejection.
UPROBE	<code>internal/gmm.HandleSecurityModeComplete</code> (<code>MacFailed</code> branch)	<code>gmm/handler.go:2297</code>	Integrity check failure on the Security Mode Complete message itself.

18 SECURITY_POLICY_VIOLATION

Type	Symbol	Location	Purpose
UPROBE	<code>internal/context.(*AmfUe).SelectSecurityAlg</code>	<code>context/amf_ue.go:603</code>	NIA0 (null integrity) is unconditionally skipped (line 610–612, hardened by commit <code>f201608</code> "fix: skip NIA0 for non-emergency NAS integrity selection"). Returns "no matched integrity algorithm" if the UE only offers NIA0. Called from <code>sm.go:285</code> ; on error, fires <code>SendRegistrationReject(Cause5GMMUESecurityCapabilitiesMismatch) + SecurityModeFailEvent</code> .

In real `free5gc` this surfaces as a Registration Reject, not a distinct message — `SECURITY_POLICY_VIOLATION` and `REG_REJECTED` share the same downstream call at this point; the distinguishing signal is the specific cause code / the fact it originated from `SelectSecurityAlg`.

19 NF_TIMEOUT

DERIVED SIGNAL

Type	Symbol	Location	Purpose
URETPROBE	(*nudmService).UeCmRegistration, SDMGetAmData	udm_service.go:382, 95	No dedicated timeout function exists — free5gc relies on the generated OpenAPI client's default HTTP timeout. Every consumer call returns a plain Go error; attach uretprobes to each and classify the returned error as context.DeadlineExceeded / net.Error.Timeout()==true.
URETPROBE	(*nausfService).SendUEAuthenticationAuthenticateRequest	ausf_service.go:51	
URETPROBE	(*npcfService).AMPolicyControlCreate	pcf_service.go:47	
URETPROBE	(*nsmfService).SendCreateSmContextRequest	smf_service.go:153	
URETPROBE	(*nssfService).NSSelectionGetForRegistration	nssf_service.go:44	

20 SLICE_REJECTED

Type	Symbol	Location	Purpose
UPROBE	internal/gmm.handleRequestedNssai	gmm/handler.go:1103	Registration-time NSSAI validation; on NSSF NSSelectionGetForRegistration failure (~1159–1165) sends Registration Reject.
UPROBE	internal/gmm.transport5GSMMessage (NSSAI check)	gmm/handler.go:83, ~161–168	Per-PDU-session S-NSSAI allowed-check (SM_CONTEXT_PENDING→SLICE_REJECTED edge), distinct from the registration-time check.
URETPROBE	(*nssfService).NSSelectionGetForRegistration	sbi/consumer/nssf_service.go:44	Literal NSSF call.

21 PDU_SESSION_REJECTED

Type	Symbol	Location	Purpose
UPROBE	internal/gmm.CreatePDUSession (reject branch)	gmm/handler.go:271–289	Same function as state 11; on SMF-side reject sends SendDLNASTransport with the SMF's N1 SM error message. Distinguish success/PDU_SESSION_REJECTED/SLICE_REJECTED by inspecting args/return at this one probe.

22 CONTEXT_SETUP_FAILED

Type	Symbol	Location	Purpose
UPROBE	internal/ngap.handleInitialContextSetupFailureMain	ngap/handler.go:1029	Exact NGAP InitialContextSetupFailure handler; cause argument classifies the NGAP failure reason.

23 CLEANUP_REQUIRED

Type	Symbol	Location	Purpose
UPROBE	internal/gmm/common.RemoveAmfUe	gmm/common/user_profile.go:13	Single real choke-point for "release partial context": releases SM contexts via SMF, deletes AM policy association via PCF, then calls ue.Remove(). Reached from every failure branch above.
UPROBE	internal/context.(*AmfUe).Remove	context/amf_ue.go:312	Low-level resource-release primitive: stops NAS timers, removes all RanUe, frees TMSI, deletes from UePool. Attach here to confirm teardown completed.

Part 3 — Supplementary Hooks & Packet-Ingress Boundary

Edge-only functions (not tied 1:1 to a state card above)

Edge	Symbol	Location	Note
NAS_SECURITY_PENDING → UDM_REGISTERING (integrity success)	<code>context.(*AmfUe).SecurityContextIsValid</code>	<code>gmm/sm.go:272</code>	Checked at SecurityMode entry; success path also visible via HandleSecurityModeComplete (2340–2346).
UE/gNB → INITIAL_CONTEXT_SETUP (setup response)	<code>internal/ngap.handlerInitialContextSetupResponse</code>	<code>ngap/handler_generated.go:4163</code>	NGAP dispatch entry, before <code>handleInitialContextSetupResponseMain</code> .
PDU_SESSION_REJECTED → UE_CONTEXT_READY (retain context)	<code>– implicit –</code>	<code>gmm/handler.go:285–289</code>	No explicit function fires; "retain" is the absence of a <code>RemoveAmfUe</code> call in the reject branch.
Any NAS message routed to GMM FSM	<code>internal/nas.Dispatch</code>	<code>nas/dispatch.go:15</code>	Fan-in point before every GMM FSM state's <code>GmmMessageEvent</code> case runs — general-purpose "message routed" probe.
FSM state transition itself	<code>free5gc/util/fsm.(*FSM).SendEvent</code>	external module (not in internal/)	Not hookable from this repo's binary section alone unless the dependency is statically linked in (it is, in the final Go binary) — symbol still resolvable via <code>go tool nm</code> .

Packet-ingress boundary for REG_RECEIVED (XDP vs. uprobe)

Layer	Symbol	Location	Note
SCTP association accept	<code>sctpListener.AcceptSCTP</code>	<code>ngap/service/service.go:94</code>	New N2 association boundary.
SCTP read loop	<code>handleConnection</code> / <code>conn.SCTPRead</code>	<code>ngap/service/service.go:183, 200</code>	* <code>sctp.SCTPConn</code> from <code>github.com/free5gc/sctp</code> , a <code>cgo/raw-socket</code> wrapper — this is the lowest Go-visible point.
Worker dispatch	<code>dispatchToWorkerPool</code>	<code>ngap/service/service.go:250</code>	Next Go-level hop after the PPID (60 = NGAP) check at line 230.
XDP (kernel, below Go)	<code>raw SCTP DATA chunk, PPID 60</code>	NIC ingress / socket filter	Strictly earlier than <code>handleConnection</code> . Only useful for raw SCTP-layer DoS/flood detection before userspace copy — not represented by any state-graph node, since decoded NGAP/NAS content (needed for real state tracking) only exists after Go-level decode.

Recommendation

Uprobes on the Go symbols in Parts 1–2 are sufficient to reconstruct all 23 conceptual states and write them into the eBPF map — they fire after NGAP/NAS decode, which is where the actual state semantics live. Reserve XDP for a separate, lower-layer signal (raw SCTP flood/DoS detection on the N2 interface) that this state graph does not model.

Quick Index — Symbol Table

#	State	Primary symbol	Location
01	REG_RECEIVED	<code>internal/nas.HandleNAS</code>	<code>nas/handler.go:14</code>
02	CONTEXT_LOOKUP	<code>(*namfService).UEContextTransferRequest</code>	<code>sbi/consumer/amf_service.go:245</code>
03	IDENTITY_PENDING	<code>gmm/message.SendIdentityRequest</code>	<code>gmm/message/send.go:91</code>
04	AUTH_VECTOR_PENDING	<code>(*nausfService).SendUEAuthenticationAuthenticateRequest</code>	<code>sbi/consumer/ausf_service.go:51</code>
05	NAS_AUTHENTICATING	<code>internal/gmm.HandleAuthenticationResponse</code>	<code>gmm/handler.go:1973</code>
06	NAS_SECURITY_PENDING	<code>internal/gmm.SecurityMode</code>	<code>gmm/sm.go:260</code>
07	UDM_REGISTERING	<code>internal/gmm.communicateWithUDM</code>	<code>gmm/handler.go:1004</code>
08	SUBSCRIPTION_LOADING	<code>(*nudmService).SDMGetAmData</code>	<code>sbi/consumer/udm_service.go:95</code>
09	POLICY_ASSOCIATING	<code>(*npcfService).AMPolicyControlCreate</code>	<code>sbi/consumer/pcf_service.go:47</code>
10	UE_CONTEXT_READY	<code>gmm/message.SendRegistrationAccept</code>	<code>gmm/message/send.go:562</code>
11	SM_CONTEXT_PENDING	<code>internal/gmm.CreatePDUSession</code>	<code>gmm/handler.go:222</code>
12	INITIAL_CONTEXT_SETUP	<code>ngap/message.SendInitialContextSetupRequest</code>	<code>ngap/message/send.go:462</code>
13	REGISTERED_CONNECTED	<code>internal/gmm.Registered</code>	<code>gmm/sm.go:65</code>
14	REG_REJECTED	<code>gmm/message.SendRegistrationReject</code>	<code>gmm/message/send.go:393</code>
15	AUTH_FAILED	<code>internal/gmm.Authentication (AuthFailEvent)</code>	<code>gmm/sm.go:222</code>
16	REPLAY_SUSPECTED	<code>ue.MacFailed (derived)</code>	<code>nas/handler.go:61</code>
17	SECURITY_FAILED	<code>nas_security.Decode</code>	<code>nas/nas_security/security.go:120</code>
18	SECURITY_POLICY_VIOLATION	<code>context.(*AmfUe).SelectSecurityAlg</code>	<code>context/amf_ue.go:603</code>

#	State	Primary symbol	Location
19	NF_TIMEOUT	consumer return-value classification (derived)	sbi/consumer/*.go
20	SLICE_REJECTED	internal/gmm.handleRequestedNssai	gmm/handler.go:1103
21	PDU_SESSION_REJECTED	internal/gmm.CreatePDUSession (reject branch)	gmm/handler.go:271-289
22	CONTEXT_SETUP_FAILED	internal/ngap.handleInitialContextSetupFailureMain	ngap/handler.go:1029
23	CLEANUP_REQUIRED	gmm/common.RemoveAmfUe	gmm/common/user_profile.go:13